

PHASE 18 / RELEASE CANDIDATE
CLOSURE

OperatorOS

Human Completion Guide

The shortest safe path from a locally verified thirteen-module ecosystem to an accepted production release.

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Scope: OperatorOS plus all thirteen active modules

Owner tasks: Replit, production PostgreSQL, DNS/TLS, providers, acceptance, and rollback

914 / 0 / 6

API PASS / FAIL / INTENTIONAL SKIP

12 / 12

EXACT-HOST BROWSER SCENARIOS

33 / 33

DATABASE RELEASE STEPS

LOCAL ECOSYSTEM: PASS | PRODUCTION: HUMAN-GATED

What is already complete

Do not repeat local restoration work unless the final merge changes runtime code. The current candidate has passed:

- 45/45 active registry, release, provider, preflight, and Replit contracts;
- database release v33/33 plan, clean apply, and idempotent reapply;
- 914 passed, 0 failed, and 6 intentional HTTP-only skips across 920 API tests;
- API, runner, and web typechecks plus the Next 15.5.22 production build;
- strict compiled supervisor health/readiness with database v33/33, SSO, and the shared worker configured;
- 12/12 local canonical-host browser scenarios across all thirteen modules;
- 2/2 compiled first-screen scenarios, including the deterministic OutCall verified-self/profile/trigger/schedule workflow and denial.

These results prove the source/local state-4 ecosystem. They do not prove that the final commit is deployed, providers are live, production data is backed up, or rollback works.

Hard stop rules

STOP ON ANY IDENTITY, READINESS, TENANT, PROVIDER-MODE, BACKUP, OR ROLLBACK FAILURE.

Stop the release immediately if any of these conditions occurs:

- the deployed Git commit differs from the reviewed `origin/main` commit;
- `/readyz` is not HTTP 200 with database release v33/33;
- the 48-check public verifier is not 48/48;
- any production-safe browser scenario fails;
- a secret, OTP, phone number, database URL, cookie, or provider signature is exposed in a log, screenshot, issue, document, or Git diff;
- a tenant can see or mutate another tenant's data;
- a module launches without its entitlement;
- Stripe, Twilio, Resend, or OpenAI is in the wrong account or mode;
- a provider callback is unsigned, points at a retired host, or cannot reject a replay/tampered request;
- the backup has no readable table-of-contents and SHA-256 checksum;
- the rollback owner, rollback commit, or restored-database switch is unknown.

Do not run child migrations, `drizzle-kit push`, ad hoc SQL directories, or the local `e2e/sso-v1.spec.ts` against production. The local test registers users and performs direct database setup. Use only the production-safe deployed 3/3 gate described below.

Release worksheet

Record values in a protected operations record, never in this repository.

Evidence	Required value
Release owner	Name and UTC start time
Reviewed Git commit	Full 40-character <code>origin/main</code> SHA
Replit deployment/build	Deployment ID and build ID

Evidence	Required value
Deployment time	UTC timestamp
Database	Provider, PostgreSQL version, and release before/after
Provider backup	Snapshot ID and UTC timestamp
Logical backup	Encrypted location, bytes, SHA-256, and TOC result
Stripe	Account ID, mode, webhook endpoint ID, safe Price IDs
Resend	Account/domain and delivered message ID
Twilio	Account SID suffix, Verify service SID, owned-line suffix
OpenAI	Project/account and successful request ID
Browser acceptance	3/3 result and UTC timestamp
OutCall acceptance	Safe provider IDs and each PASS/FAIL result
Rollback	Last-known-good commit and restored-database target

Step 1 - Freeze the exact candidate

Run from **C:\Dev\OperatorOS** in PowerShell:

```
git fetch --prune origin
git switch main
git pull --ff-only origin main
$candidate = (git rev-parse HEAD).Trim()
if ($candidate -ne (git rev-parse origin/main).Trim()) {
    throw 'Local main does not match origin/main'
}
git status --short

$env:CI = 'true'
corepack pnpm install --frozen-lockfile
corepack pnpm typecheck
$env:INTERNAL_API_URL = 'http://localhost:5001'
corepack pnpm build:production
corepack pnpm db:plan
```

Require a clean worktree, release version 33, 33 ordered non-destructive steps, and final step **outcall_product_operations**. The deployment must later expose the same **\$candidate** through health and readiness.

Step 2 - Resolve the remaining account decisions

Complete these owner decisions before entering live values:

- 1 Stripe mode:** **.replit** currently documents **STRIPE_MODE=test**. Keep it only for a sandbox launch. A revenue-live promotion requires explicit owner approval, reviewed live keys/prices, a low-risk live transaction and refund, and a source/config change that makes production mode intentionally **live**.
- 2 Legacy add-on prices:** reconfirm **STRIPE_PRICE_ADDON_NINJA_LAUNCH_KIT** and **STRIPE_PRICE_ADDON_STUDYFORGE_AI**. The prior handoff reported old live-mode IDs that fail under the sandbox key. Replace them with correct same-mode \$49/month prices, or remove them so checkout fails closed.
- 3 OutCall commercial price:** create and set **STRIPE_PRICE_ADDON_OUTCALL** only when OutCall is intentionally purchasable. Entitled accounts can launch the module without pretending checkout is ready.

- 4 **OpenAI quota:** fund the intended OpenAI project/account and rerun the AI profile. A present key that returns `insufficient_quota` is not AI-ready.
- 5 **Twilio authority:** use the intended production account and owned numbers. The Replit connector may support CallCommand, but signed webhooks and OutCall require the primary `TWILIO_AUTH_TOKEN` in the deployment secret manager.

Step 3 - Configure Replit Publishing

Enter secrets only in **Publishing -> Edit Commands and Secrets**. Development workspace values do not prove that the published deployment received them.

Core required values

- `DATABASE_URL`
- `SESSION_SECRET` (high entropy, 24+ characters)
- `SSO_CODE_ENCRYPTION_SECRET` (independent, high entropy, 32+ characters)
- `APP_ENV=production`
- `NODE_ENV=production`
- `OPERATOROS_BASE_URL=https://operatoros.net`
- `OPERATOROS_APPS_URL=https://app.operatoros.net/`
- `INTERNAL_API_URL=http://localhost:5001`
- `OPERATOROS_DATABASE_RELEASE_MODE=apply`
- `TRUST_PROXY=true`
- `OPERATOROS_BOOTSTRAP_SUPER_ADMIN_EMAIL=john@shotgunninja.com`
- all thirteen canonical module URL values from `config/production-environment.contract.json`

Leave `APP_URL`, `COOKIE_DOMAIN`, `NEXT_PUBLIC_API_URL`, `OPERATOROS_DATABASE_RELEASE_APPLIED`, `OUTCALL_TEST_ADAPTER`, and `CALLCOMMAND_TEST_ADAPTER` unset. Leave `ALLOW_LEGACY_SSO_ROLLBACK` and `ALLOW_UNSAFE_COMMANDS` unset or false. Never set a parent-domain session cookie.

Revenue and email

- Stripe secret key, webhook signing secret, intentional `STRIPE_MODE`, five shared plan/seat Price IDs, and each intentionally sellable add-on Price ID;
- `RESEND_API_KEY` and the verified Shotgun Ninjas sender variable used by the deployment;
- verify that `shotgunninja.com` remains verified in the selected Resend account.

CallCommand

- a bound Replit Twilio connector or `TWILIO_ACCOUNT_SID`, `TWILIO_AUTH_TOKEN`, and `TWILIO_FROM_NUMBER`;
- `TWILIO_PUBLIC_BASE_URL=https://callcommand-ai.operatoros.net`.

OutCall

- `OUTCALL_PUBLIC_URL=https://outcall.operatoros.net`
- independent versioned `OUTCALL_FIELD_ENCRYPTION_KEY` and `OUTCALL_LOOKUP_HMAC_KEY` (32+ characters each; never reuse session/SSO keys)
- `TWILIO_ACCOUNT_SID`
- `TWILIO_AUTH_TOKEN`
- optional outbound API-key pair `TWILIO_API_KEY_SID` and `TWILIO_API_KEY_SECRET`

- `TWILIO_VERIFY_SERVICE_SID`
- `TWILIO_PHONE_NUMBER` (approved owned E.164 line)
- `TWILIO_ALLOWED_COUNTRIES=US, CA` or the narrower approved controlled set
- `OUTCALL_LIVE_PROVIDER` absent/disabled until the controlled provider gate; set exactly `enabled` only for that accepted activation step

Configure the Twilio inbound SMS callback exactly as:

`https://outcall.operatoros.net/api/modules/outcall/webhooks/twilio/sms`

OperatorOS generates the voice status and gather callbacks for each controlled call. Do not replace them with a generic or standalone-domain callback.

AI

- `OPENAI_API_KEY`
- reviewed model/project variables already documented by the repository

Step 4 - Run production preflight

Run profiles separately first so failures have one owner:

```
corepack pnpm preflight:production -- --core
corepack pnpm preflight:production -- --revenue-ready
corepack pnpm preflight:production -- --email-ready
corepack pnpm preflight:production -- --callcommand-ready
corepack pnpm preflight:production -- --ai-ready
```

Before OutCall activation, the first five profiles must pass. After the controlled Twilio configuration is reviewed and `OUTCALL_LIVE_PROVIDER=enabled` is deliberately set, run:

```
corepack pnpm preflight:production -- --outcall-ready
corepack pnpm preflight:production -- --all
```

Require PASS for `core`, `revenue`, `email`, `callcommand`, `outcall`, and `ai`. The preflight prints variable names but must not print values.

Step 5 - Back up production before release v33

- 1 Capture a provider-managed encrypted snapshot and confirm its ID/status.
- 2 From a trusted operator workstation with `DATABASE_URL` supplied by the secret manager, create the encrypted logical archive:

```
pg_dump --format=custom --no-owner --no-acl --file operatoros.dump $env:DATABASE_URL
pg_restore --list operatoros.dump
Get-FileHash -Algorithm SHA256 -LiteralPath operatoros.dump
```

- 3 Record UTC time, commit, PostgreSQL version, byte size, duration, protected location, SHA-256, and readable TOC result.
- 4 Prefer a restore rehearsal into a new isolated database with all providers disabled. Do not restore over production in place.

Step 6 - Deploy the exact merged commit

In the OperatorOS Replit autoscale deployment:

- 1 select the reviewed `$candidate` from `origin/main`;

- 2 confirm the checked-in build command performs a frozen pnpm install and `pnpm build:production`;
- 3 confirm the run command is `node scripts/start-unified-runtime.mjs`;
- 4 start the deployment and watch the ordered v33 release;
- 5 require the log message `Fastify ready; starting Next` before public Next starts;
- 6 record Replit deployment ID, build ID, Git commit, and UTC timestamp.

The supervisor owns the production v33 apply. Do not separately run an ad hoc production `db:apply` while the deployment is starting.

Step 7 - Prove public identity and all 17 hosts

Set the commit pin and run the read-only verifier:

```
$env:OPERATOROS_EXPECTED_RELEASE_COMMIT = $candidate
corepack pnpm verify:production
```

Require 48/48. `/api/health` and `/readyz` must expose the same exact commit, build identity, valid timestamps, and database release v33/33.

Confirm valid TLS and expected host behavior for:

- `operatoros.net`, `app.operatoros.net`, `auth.operatoros.net`, `api.operatoros.net`;
- `tradeflowkit`, `torqueshed`, `techdeck`, `pulsedesk`, `faultlinelab`, `ninja-pool-hall`, `brandforgeos`, `snapproofos`, `studyforge-ai`, `ninjalaunchkit`, `callcommand-ai`, `ninjamotion`, and `outcall` under `.operatoros.net`.

The Replit alias is not an SSO callback, CORS origin, or return target.

Step 8 - Run the production-safe authenticated 3/3 gate

Provision two synthetic accounts in the intended production tenant system:

- one active tenant entitled to all thirteen modules;
- one active tenant deliberately denied both TechDeck and OutCall.

Load the six values below from a protected secret manager into the current PowerShell process. Do not record them in shell history or screenshots.

```
$env:E2E_PHASE17_EMAIL = '<entitled acceptance user>'
$env:E2E_PHASE17_PASSWORD = '<secret>'
$env:E2E_PHASE17_TENANT_ID = '<entitled tenant UUID>'
$env:E2E_PHASE17_DENIED_EMAIL = '<denied acceptance user>'
$env:E2E_PHASE17_DENIED_PASSWORD = '<secret>'
$env:E2E_PHASE17_DENIED_TENANT_ID = '<tenant UUID denied TechDeck and OutCall>'
Remove-Item Env:E2E_PRODUCTION_HOSTS -ErrorAction SilentlyContinue
corepack pnpm --dir apps/web test:e2e:phase17-deployed
```

The script name is historical; its current contract requires thirteen enabled modules and active OutCall denial. Require 3/3:

- 1 one login launches every enabled module and global logout revokes siblings;
- 2 TechDeck local logout preserves the PulseDesk sibling session;
- 3 the denied tenant receives `MODULE_ACCESS_DENIED` for TechDeck and OutCall with no handoff URL.

Clear all six `E2E_PHASE17_*` variables after the run.

Step 9 - Execute controlled provider acceptance

Use synthetic, owner-controlled data only. Record provider IDs in the protected evidence record, not in Git.

Resend

- send one invite to a controlled non-customer mailbox;
- confirm delivery and accept the invite through normal review;
- record the safe message ID and final PASS/FAIL.

OpenAI

- after funding the intended project, run one bounded shared AI generation;
- require a real successful provider response and one usage event;
- confirm no prompt/customer secret appears in logs;
- record the request ID and PASS/FAIL.

Stripe

- first repeat sandbox checkout/webhook/cancel/duplicate/tamper checks using same-mode Price IDs;
- for live cutover, obtain explicit owner approval for one low-risk transaction and refund;
- verify entitlement grant, seat math, tenant isolation, webhook signature, idempotency, cancellation, and refund reconciliation;
- immediately stop if any test/live mode mismatch occurs.

CallCommand

- confirm the owned line and signed callback base;
- place only the approved controlled consent-first call;
- verify consent, suppression, callback signature/replay handling, disposition, and recording-off behavior.

OutCall verified-self sequence

Set `OUTCALL_LIVE_PROVIDER=enabled` only when ready to run this exact sequence:

- 1 accept the non-emergency safety boundary;
- 2 send one Verify challenge to the owner's controlled +1 number;
- 3 submit the OTP through the product; never record the OTP;
- 4 create one neutral profile and private exact-match trigger;
- 5 schedule one immediate verified-self call;
- 6 acknowledge by DTMF and confirm signed status/gather callbacks;
- 7 send one private trigger SMS to the owned OutCall number;
- 8 replay one captured provider callback and require duplicate-ignore behavior;
- 9 tamper with a callback signature and require rejection;
- 10 schedule then cancel one delayed request and confirm no call occurs;
- 11 download the private export and confirm masking/protected fields;
- 12 use a separate disposable acceptance identity to test password-confirmed `DELETE OUTCALL`, confirming central audit and billing usage remain.

OutCall must never call an arbitrary destination, record audio, impersonate a person, claim emergency dispatch, expose a trigger phrase, or broaden beyond the approved verified-self boundary. If any provider step fails, set **OUTCALL_LIVE_PROVIDER** back to disabled/absent and keep the module visible but provider actions fail-closed.

Step 10 - Promotion decision and rollback proof

Promote only when all of these are true on one exact deployed commit:

- production preflight PASS for every marketed provider profile;
- provider snapshot and logical backup accepted;
- supervisor release v33/33 and readiness PASS;
- public verifier 48/48;
- production-safe authenticated browser 3/3;
- controlled Resend, OpenAI, Stripe, CallCommand, and OutCall scopes either PASS or are explicitly disabled and not marketed as live;
- rollback target and owner are recorded.

Application rollback: redeploy the last-known-good reviewed commit, pin **OPERATOROS_EXPECTED_RELEASE_COMMIT** to it, and rerun the public verifier.

Database rollback: restore the accepted archive into a **new** database, run the supported release/readiness checks with providers disabled, then switch **DATABASE_URL**/traffic after review. Never overwrite the production database in place. Release v33 is additive; an application rollback normally leaves its tables in place unless a verified restore is required.

Final sign-off

Gate	Owner initials	UTC	Result/evidence reference
Exact merged commit/build			
Core/provider preflight			
Snapshot and logical backup			
v33/33 supervisor readiness			
Public verifier 48/48			
Deployed browser 3/3			
Resend			
OpenAI			
Stripe			
CallCommand			
OutCall			
Rollback proof			

Final decision: PROMOTE / HOLD / ROLLBACK

Owner signature: _____

UTC timestamp: _____